

HackMan3D Orbit Controller

Assembly Guide

1. Preparing the Hall Effect Joysticks

Bend the two side mounting tabs downward slightly (do not flatten them completely). Cut off the third mounting tab opposite the connector pins. Bend the five connector pins outward until they are horizontal.

2. Wiring the Hall Effect Joysticks

Cut one female connector from each jumper wire. **Before soldering, slide a piece of heat-shrink tubing onto every wire.** Once soldered, the remaining connector may be too large to pass through the tubing. Solder the wires exactly as shown in the wiring diagram. **Keep the recommended wire colors.** This makes the final assembly and troubleshooting much easier. Move the heat-shrink tubing over each solder joint and shrink it. **Important:** Forgetting the heat-shrink tubing means you may have to desolder the wire.

3. Preparing the Knob

Before installing the joysticks into the Electronics Carrier, press the small **TPU printed balls** onto each joystick shaft. These must be installed before the joysticks are mounted.

4. Button Module

If using the optional mechanical buttons, solder the wires according to the wiring diagram. Install the switches, connect the common ground after the switches are fitted, route the wires through the dedicated holes and secure the Button Module using **2 × M3x8** screws.

5. Electronics Carrier

Install the four prepared Hall Effect joysticks and the Arduino Pro Micro onto the Electronics Carrier. Route all wires through the dedicated channels before placing the assembly into the Base. Secure the Electronics Carrier into the Base using **4 × M3x6** screws.

6. Lower Knob Assembly

Once all four joysticks are installed, assemble the **Knob Lower Base** and **Knob Lower Cap** around the TPU balls. Secure both parts together using **4 × M2x10 countersunk screws**.

7. Top Cover

Install the Top Cover using **6 × M2x6 countersunk screws**.

8. Upper Knob Assembly

Place the **Knob Grip (TPU)** between the **Knob Upper Base** and **Knob Upper Cap**. Secure the three parts together using **4 × M3x10** screws. Finally, attach the complete knob assembly using **1 × M3x12** screw.

9. Final Steps

Install the keycaps, connect the controller with a USB-C cable, install the NavCore board package, then upload the firmware from the HackMan3D GitHub repository.

Hardware Summary

M3 Screws

4 × M3x6
2 × M3x8
4 × M3x10
1 × M3x12

M2 Countersunk Screws

4 × M2x10
6 × M2x6